meta-pipe: An LLM-agent pipeline for end-to-end automated systematic review and meta-analysis

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Abstract

Objective: To describe the architecture and design rationale of meta-pipe, an open-source large language model (LLM)-agent pipeline that integrates the complete systematic review and meta-analysis (SR/MA) workflow — from literature search through statistical analysis, manuscript generation, and quality assurance — with mandatory human oversight at critical decision points.

Study Design and Setting: We developed a 10-stage modular pipeline integrating Claude (Anthropic; Opus 4 for reasoning, Haiku 3.5 for classification) for LLM-assisted screening and extraction, Python (~3,600 lines of code) for automation, R (meta, metafor, gemtc, netmeta) for statistical analysis, and Quarto for manuscript rendering. Five mandatory human decision points enforce oversight. We systematically compared meta-pipe's capabilities with five existing SR automation tools based on published documentation as of March 2026.

Results: meta-pipe offers four capabilities not available in any single existing tool: automated manuscript generation from analysis outputs, semi-automated GRADE assessment, overclaim detection (12 predefined patterns), and dual-paradigm network meta-analysis (Bayesian and frequentist). Estimated API cost is $15–30 per typical 5–10 study review. No validation data are reported; this is a system description, not a validation study.

Conclusion: End-to-end AI-assisted evidence synthesis is architecturally feasible as an open-source tool with mandatory human oversight. Formal validation reproducing published Cochrane reviews is underway and essential before routine use.

Keywords: systematic review, meta-analysis, large language model, automation, evidence synthesis, artificial intelligence

1. Introduction

Systematic reviews and meta-analyses (SR/MA) are resource-intensive: traditional reviews require approximately 1,100 person-hours [1, 2] and 67 weeks [3], with the most efficient reported workflow requiring 61 person-hours [4].

Several automation tools address this burden. Covidence extends screening to extraction [5]; TrialMind combines search, screening, and extraction [6]; otto-SR automates screening through meta-analysis [7]; and Nested Knowledge covers SR through network meta-analysis [8]. A scoping review of 37 studies found LLM approaches covered 10 of 13 SR steps but concluded they are "not yet ready for standalone use" [9]. The three uncovered steps were protocol development, certainty of evidence assessment, and interpretation — the latter two of which meta-pipe addresses.

Despite this progress, no open-source tool integrates statistical analysis, manuscript generation, and quality assurance as a unified pipeline. To address this gap, we developed meta-pipe, an open-source LLM-agent pipeline orchestrating the complete SR/MA workflow across 10 stages with five mandatory human

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decision points. As a system description — not a validation study — this paper contributes: (1) the pipeline architecture with standardized data contracts; (2) automated manuscript generation from meta-analytic outputs; (3) a quality assurance layer (GRADE assessment and overclaim detection); and (4) a systematic comparison with five existing tools. Formal validation reproducing published Cochrane reviews is underway separately.

2. Methods

2.1. Pipeline Architecture

meta-pipe comprises 10 sequential stages (Table 1), each implemented as a self-contained skill module with defined inputs, outputs, and quality thresholds. Each skill module is a structured prompt document containing workflow guidance, command references, validation criteria, and input/output specifications (Supplementary S2). Inter-stage data contracts specify file schemas at each boundary: BibTeX for bibliography records, CSV for screening and extraction data, JSON for configuration, and Quarto markdown for manuscript components (Supplementary S3). Schema validation is currently documentation-based; programmatic enforcement is planned.

Human decision points are embedded at five junctures: (1) PICO definition and eligibility criteria; (2) screening disagreement resolution; (3) analysis type selection (pairwise vs network meta-analysis); (4) GRADE quality assessment; and (5) interpretation and clinical implications. These gates retain human judgment where domain expertise is essential, consistent with consensus that AI should augment rather than replace human reviewers [9, 10].

2.2. Technology Stack

The pipeline integrates four technologies: (a) Claude [11] (Anthropic) for LLM orchestration — using Opus 4 with extended thinking for complex reasoning tasks (screening adjudication, risk-of-bias assessment, manuscript generation) and Haiku 3.5 for high-throughput simple tasks (title/abstract classification, data validation) — introducing a commercial API dependency that affects reproducibility, cost, and availability; portability to other LLMs (GPT-4, Gemini, Llama) has not been tested; prompt versioning is partially implemented; (b) Python (˜3,600 lines of code) for data processing, LLM-assisted screening and extraction, validation, and quality assurance, managed via uv with lock files; (c) R for statistical analysis: meta v8.0 [12], metafor v4.8 [13], gemtc v1.0 [14], and netmeta v3.4 [15], with package versions locked via renv; and (d) Quarto for manuscript rendering to HTML, DOCX, and PDF with validated cross-references and citations.

2.3. Search and Screening (Stages 01–03)

Literature searches use PubMed and Scopus by default, with optional extension to Embase and the Cochrane Library. The default selection does not include CENTRAL or Embase, standard requirements for Cochrane-compliant reviews; this is acknowledged as a limitation. Search strategies are LLM-drafted based on the PICO definition and require mandatory expert review and refinement before execution.

Title and abstract screening employs a test-retest reliability approach: the LLM performs two independent screening passes at temperature 0.3, and agreement is quantified using Cohen's kappa [16]. We use "test-retest reliability" rather than "inter-rater agreement" because both passes use the same model with the same training data; this measures screening decision stability under stochastic decoding variation, not independent assessment quality. The minimum kappa threshold is 0.60 (moderate agreement per Landis and Koch), below Cochrane's 0.80 for dual review. Because test-retest kappa measures intra-model stability rather than inter-rater reliability, the 0.80 benchmark is not directly applicable: kappa below 0.60 at temperature 0.3 indicates prompt ambiguity or insufficiently specific PICO criteria requiring human review. This approach does not substitute for independent dual review as recommended by PRISMA 2020.

2.4. Data Extraction and Risk of Bias (Stages 04–05)

Full-text retrieval uses Unpaywall API integration with manual fallback. PDF documents are processed by passing full text to the LLM context window; the pipeline cannot reliably extract from figures, supplementary materials, or complex multi-page tables. Data extraction employs structured templates with automated validation (range checks, consistency verification, confidence scoring). Confidence scores are LLM-generated and uncalibrated; they serve as triage rather than validated accuracy metrics. Specific failure modes include extracting from wrong tables, confusing intention-to-treat and per-protocol populations, and misidentifying primary endpoints. This deviates from PRISMA 2020 Item 11d, which recommends at least two independent extractors.

Risk of bias is assessed using RoB 2 (randomized trials) or ROBINS-I (observational studies) with LLM-assisted domain-level judgment and mandatory human review.

2.5. Statistical Analysis (Stage 06)

Pairwise meta-analyses use REML random-effects models with Hartung-Knapp adjustment for confidence intervals. Heterogeneity is quantified using I^2, Cochran's Q, and prediction intervals. Publication bias is assessed using funnel plots, Egger's/Peters' tests, and trim-and-fill analysis [17]; the pipeline warns when k < 10. Sensitivity analyses include leave-one-out analysis and influence diagnostics.

Network meta-analysis uses a dual-paradigm approach: Bayesian (gemtc [14] with MCMC) and frequentist (netmeta [15]). Discordance — defined as reversed top-three treatment rankings or point estimates differing by >0.2 SMD or odds ratio ratio >1.5 — triggers closer examination. These are pragmatic author-chosen thresholds informed by conventional effect size benchmarks, not formally calibrated.

2.6. Manuscript Generation and Quality Assurance (Stages 07–09)

Manuscript assembly uses Quarto templates for each IMRaD (Introduction, Methods, Results, and Discussion) section, reading effect estimates directly from R output files to prevent hallucination of statistical results. This capability is unique among current SR automation tools.

Quality assurance comprises PRISMA 2020 checklist validation (27 items), semi-automated GRADE assessment (with mandatory human review), and automated overclaim detection scanning for 12 predefined patterns of unsupported claims (Supplementary S4). The pipeline implements stage-level checkpointing for re-execution from any failure point; API failures trigger exponential-backoff retry for up to three attempts.

3. Results

3.1. Pipeline Implementation

The pipeline comprises 12 skill modules (10 primary stage modules plus 2 utility modules for project setup and configuration), 24 Python automation scripts (~3,600 lines of code), 5 R analysis scripts, and 74 methodology reference documents. The source code is available under the MIT license (repository URL and Zenodo DOI to be added upon acceptance).

3.2. Unique Capabilities

Four capabilities distinguish meta-pipe from existing tools (Table 2). First, *manuscript generation*: meta-pipe is the only SR automation tool producing structured IMRaD manuscripts directly from analysis outputs, with effect estimates read from R files to prevent hallucination. Second, *GRADE assessment*: semi-automated domain-level suggestions (risk of bias, inconsistency, indirectness, imprecision, publication bias) requiring mandatory human adjudication; no other tool offers this. Third, *overclaim detection*: automated scanning for 12 predefined patterns of unsupported claims, though precision and recall have not been formally evaluated. Fourth, *dual-paradigm NMA*: Bayesian (gemtc) and frequentist (netmeta) concordance checks, tested with synthetic data but not yet exercised on a real clinical review.

3.3. Feature Comparison

Table 2 compares meta-pipe with five existing SR/MA tools based on published documentation as of March 2026. Each tool excels within its scope; meta-pipe's positioning is best understood through its unique contributions — open-source end-to-end coverage, manuscript generation, quality assurance, and GRADE — rather than breadth claims. We do not claim superiority in screening or extraction accuracy, as we have no component-level validation data. otto-SR's reproduction of 12 Cochrane reviews [7] represents a validation standard that meta-pipe has not yet achieved.

3.4. Cost Estimate

API costs are estimated at $15–30 per typical 5–10 study pairwise meta-analysis. These are theoretical projections based on Claude API token pricing, not empirical measurements. The approximate per-stage breakdown is: screening ~$2–5, extraction ~$8–18 (driven by full-text PDF processing), risk-of-bias ~$2–4, and manuscript generation ~$1–3. Statistical analysis incurs minimal LLM costs. Cost scaling beyond 50 studies has not been characterized; per-stage token logging is planned for the validation study.

4. Discussion

4.1. Principal Findings

This paper describes the architecture and design rationale of meta-pipe, an open-source pipeline covering all 10 SR/MA stages. The contribution is architectural, not empirical: we have not demonstrated that the pipeline produces accurate outputs; that evidence must come from the planned Cochrane reproduction study. What we have shown is that end-to-end AI-assisted evidence synthesis is technically feasible as an open-source tool with mandatory human oversight.

4.2. Comparison with Existing Tools

meta-pipe complements rather than competes with existing tools (Table 2). Its core differentiators — open-source end-to-end scope, manuscript generation, GRADE assessment, and overclaim detection — address capabilities not available in any single existing tool. Users can achieve similar coverage by chaining tools (e.g., Covidence for screening, R for analysis, manual manuscript writing), but meta-pipe's value lies in integrated data flow and quality gates that reduce manual transfer steps.

Concurrent with our work, Cao et al. presented otto-SR, an agentic LLM pipeline that reproduced 12 Cochrane reviews with 96.7% screening sensitivity and 93.1% extraction accuracy [7]. Where otto-SR optimizes for full automation, meta-pipe targets de novo synthesis with human decision points, additionally covering NMA, manuscript generation, and GRADE. The critical difference is evidentiary: otto-SR has demonstrated that its pipeline works; meta-pipe has described an architecture that could work. Several caveats apply: otto-SR is a preprint using LLM-as-judge evaluation and does not report cost data; however, its quantitative concordance metrics represent a validation standard meta-pipe has not achieved.

4.3. Human Oversight and Reproducibility

The five mandatory human gates align with the Cochrane 2025 AI position statement [10] and developing RAISE guidelines [18]. We acknowledge a tension: otto-SR's results suggest full automation can outperform humans on individual tasks. The human gates are thus better understood as trust-building mechanisms for novel research questions — where no gold-standard review exists — rather than proven quality improvements. Whether human oversight improves or degrades overall pipeline accuracy is an empirical question for the validation study.

Reproducibility is supported by renv (R) and uv (Python) lock files, but LLM outputs are inherently stochastic and model deprecation by commercial providers remains unresolved. The "open source" claim warrants qualification: pipeline code is public, but the Claude API backbone is proprietary; portability to alternative LLMs has not been tested.

4.4. Error Propagation

In a 10-stage sequential pipeline, errors compound: a screening false negative eliminates studies from all downstream analyses. Under naïve independence assumptions, 95% per-stage accuracy yields only 60% end-to-end — though this substantially overstates the problem because human decision points correct errors, many errors are non-compounding (a misclassified study is simply absent rather than introducing correlated downstream errors), and R-based statistical stages do not introduce LLM errors. Formal error propagation analysis — empirical or theoretical — is needed to characterize reliability.

4.5. Limitations

Methodological: no validation data (the essential next step); LLM screening is test-retest, not PRISMA 2020 dual review; extraction confidence scores are uncalibrated; no systematic failure mode documentation. *Technical*: commercial API dependency affecting cost, privacy, and reproducibility; partially implemented prompt versioning; untested LLM portability; NMA and overclaim detection not validated on real clinical reviews. *Scope*: programming skill barrier (R, Python, command line); English-language literature only; default databases (PubMed, Scopus) exclude CENTRAL and Embase, falling short of Cochrane search requirements.

4.6. Conclusions

meta-pipe demonstrates that open-source, end-to-end AI-assisted SR/MA is architecturally feasible when designed with mandatory human oversight and automated quality gates. Its unique capabilities — manuscript generation from analysis outputs, semi-automated GRADE assessment, overclaim detection, and dual-paradigm NMA — are not available in any single existing tool. Formal validation reproducing published Cochrane reviews is the essential next step. We invite independent groups to validate meta-pipe and encourage the development of interoperability standards that would allow users to combine the strongest components of different SR automation tools.

Tables

4.7. Table 1. meta-pipe Pipeline Stages

Table 1: meta-pipe pipeline stages with key tasks, outputs, automated quality gates, and human decision points. Stage 03b is a sub-stage of the screening phase.

Stage	Module	Key Tasks	Outputs	Quality Gate	Human Decision
00	Topic intake	Brainstorming, feasibility assessment	PICO definition, eligibility criteria	Completeness check	Yes: define scope
01-02	Search & bibliography	Database search, deduplication	dedupe.bib, search strategy	Duplicate detection	No
03	Screening	LLM-assisted screening with test-retest check	decisions.csv, kappa statistic	Kappa $\geq$ 0.60	Yes: resolve disagreements
03b	Analysis gate	Pairwise vs NMA determination	analysis-type-decision.md	Network geometry check	Yes: confirm analysis type

Stage	Module	Key Tasks	Outputs	Quality Gate	Human Decision
04	Full-text management	PDF retrieval, Unpaywall integration	manifest.csv, full-text PDFs	Retrieval rate	No
05	Data extraction	LLM-assisted extraction, validation	extraction.csv, data dictionary	Range/consistency checks	Yes: review flagged items
06	Statistical analysis	Meta-analysis (pairwise or NMA)	Forest plots, effect estimates	Model convergence	No
07	Manuscript	Quarto-based assembly, rendering	IMRaD manuscript (HTML/PDF/DOCX)	Word count, structure	No
08	Peer review	GRADE assessment, SoF tables	GRADE profiles, reviewer report	Domain completeness	Yes: quality judgments
09	Quality assurance	PRISMA validation, overclaim detection	QA report	Pattern match count	No

4.8. Table 2. Feature Comparison with Existing Tools

Figures

4.9. Figure 1. meta-pipe Pipeline Architecture

Supplementary Materials

4.10. S1. Representative Prompt Templates

4.10.1. S1.1 Screening Prompt Template

The following is a representative prompt template used for title and abstract screening (Stage 03). The actual prompt is parameterized with the PICO definition and eligibility criteria defined in Stage 00.

```
You are a systematic review screening assistant. Your task is to evaluate
whether a study meets the inclusion criteria for a systematic review.

**Inclusion Criteria:**
- Population: {population}
- Intervention: {intervention}
- Comparator: {comparator}
- Outcomes: {outcomes}
- Study design: {study_design}
- Date range: {date_range}

**Instructions:**
1. Read the title and abstract below.
2. Classify as INCLUDE, EXCLUDE, or UNCERTAIN.
3. Provide a brief rationale (1-2 sentences).
4. Rate your confidence: HIGH, MODERATE, or LOW.

If the abstract lacks sufficient information to determine eligibility,
```

Table 2: Feature comparison of meta-pipe with existing systematic review automation tools. **Bold AI** indicates AI-assisted automation; plain text indicates supported but manual or using traditional methods. Dash (-) indicates feature not available. Based on published documentation and peer-reviewed publications as of March 2026. [a]Covidence extraction AI is partial (select fields). [b]DistillerSR provides risk-of-bias templates and forms but not AI-assisted assessment. [c]otto-SR's preprint does not report NMA capability; absence from the preprint does not confirm absence of the feature. [d]Implemented but not yet validated on real clinical reviews. [e]Validation study reproducing Cochrane reviews is underway.

Dimension	Covidence	TrialMind	otto-SR	Nested Knowledge	DistillerSR	**meta-pipe**
Intended scope	SR mgmt	Search→ Extract	Screen→ Extract+RoB	Full SR/MA	SR mgmt	Full SR/MA
Search strategy	-	**AI**	-	**AI**	-	**AI**
T/A screening	**AI**	**AI**	**AI**	**AI**	**AI**	**AI**
Full-text screening	Manual	**AI**	**AI**	Manual	Manual	**AI**
Data extraction	**AI**[a]	**AI**	**AI**	**AI**	**AI**	**AI**
Risk of bias	Template	-	**AI**	Template	Template[b]	**AI**
Pairwise MA	-	-	R (metafor)	**AI**	-	R (meta/metafor)
Network MA	-	-	-[c]	**AI**	-	R (gemtc/ netmeta)[d]
Manuscript generation	-	-	-	-	-	**AI** (Quarto)
GRADE / QA	-	-	-	-	-	Semi-auto
Overclaim detection	-	-	-	-	-	**AI**[d]
Open source	No	Unclear	No	No	No	**Yes**
Validation evidence	Internal	Trial-ReviewBench	12 Cochrane (preprint)	Published	Internal	None[e]

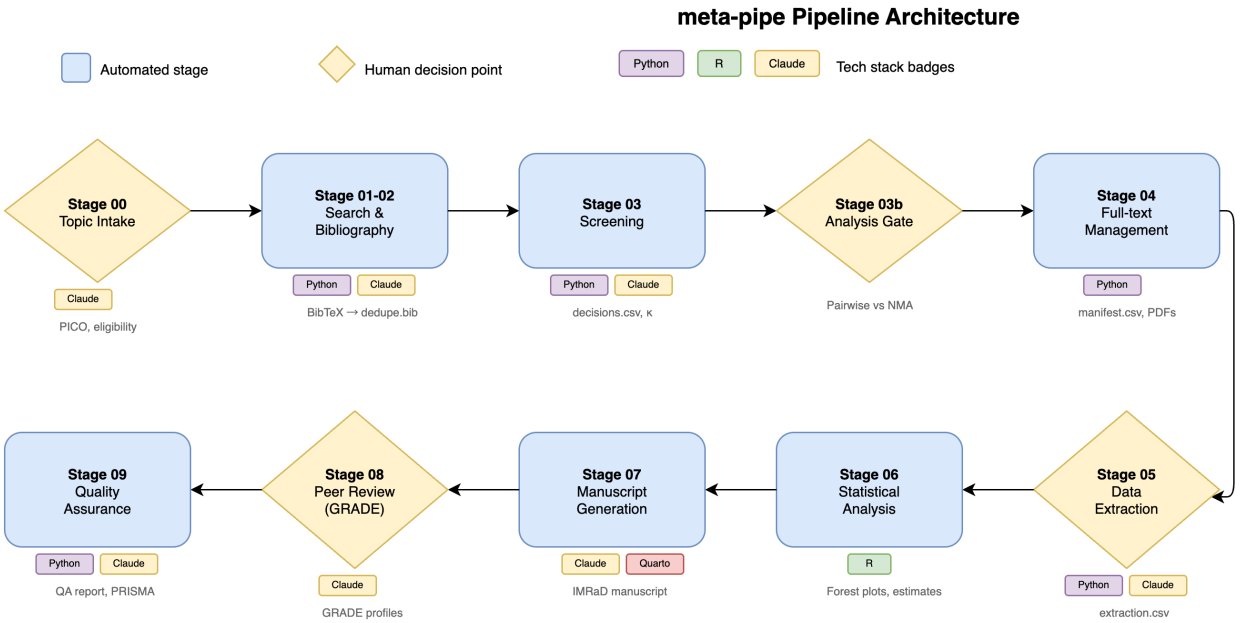


Figure 1: meta-pipe pipeline architecture. The 10-stage workflow proceeds from topic intake (Stage 00) through quality assurance (Stage 09). Blue boxes indicate automated stages; orange diamonds indicate mandatory human decision points. Arrows represent data flow between stages with standardized file formats (BibTeX, CSV, JSON, Quarto markdown) at each boundary. The pipeline integrates Claude AI (LLM orchestration), Python (automation), R (statistical analysis), and Quarto (manuscript rendering).

```
classify as UNCERTAIN (not EXCLUDE).

**Title:** {title}
**Abstract:** {abstract}

**Response format (JSON):**
{{"decision": "INCLUDE|EXCLUDE|UNCERTAIN",
  "rationale": "...",
  "confidence": "HIGH|MODERATE|LOW"}}
```

4.10.2. S1.2 Data Extraction Prompt Template

```
You are a data extraction assistant for systematic reviews. Extract the
following fields from the full-text article provided.

**Required fields:**
- study_id: First author last name + year (e.g., "Smith2024")
- study_design: RCT, cohort, case-control, cross-sectional, etc.
- n_total: Total number of participants randomized/enrolled
- n_intervention: Number in intervention group
- n_control: Number in control group
- intervention_description: Detailed intervention description
- control_description: Detailed control description
- primary_outcome: Name and definition of primary outcome
- effect_estimate: Point estimate (RR, OR, HR, MD, etc.)
- ci_lower: Lower bound of 95% confidence interval
- ci_upper: Upper bound of 95% confidence interval
```

```
- p_value: P-value if reported
- follow_up_duration: Duration and unit
- country: Study country/countries
- funding_source: Funding information

**Instructions:**
- Extract values exactly as reported in the paper.
- If a field is not reported, enter "NR" (not reported).
- For each field, rate confidence as HIGH, MODERATE, or LOW.
- Flag any values that seem inconsistent or require verification.
- Do NOT infer, calculate, or impute values not explicitly stated.

**Article text:**
{article_text}

**Response format (JSON):**
{{"study_id": "...", "confidence": "...", ...}}
```

4.11. S2. Skill Module Structure

Each skill module is a markdown document with the following structure:

```
# Stage [N]: [Stage Name]

## Overview
[Brief description of the stage's purpose and role in the pipeline]

## Prerequisites
- Required input files and their schemas
- Required software and package versions
- Human decisions from prior stages

## Workflow Steps
1. [Step 1: description]
   - Command: `python scripts/[script_name].py --input [file] --output [file]`
   - Expected output: [description]
   - Validation: [check to perform]
2. [Step 2: description]
   ...

## Quality Gates
- [Gate 1]: [threshold and action if not met]
- [Gate 2]: [threshold and action if not met]

## Human Decision Point (if applicable)
- **Trigger**: [condition]
- **Required action**: [what the human must decide]
- **Documentation**: [how the decision is recorded]

## Output Specification
- File: [filename and format]
- Schema: [column headers or JSON structure]
- Validation: [automated checks before passing to next stage]
```

```
## Troubleshooting
- [Common issue 1]: [resolution]
- [Common issue 2]: [resolution]
```

4.12. S3. Inter-Stage Data Contracts

Stage Boundary	Format	Key Fields / Schema	Validation
00 → 01	JSON	`pico.json`: population, intervention, comparator, outcomes, study_design, date_range, databases	Documentation only
01 → 02	BibTeX	Standard BibTeX fields (author, title, year, journal, abstract, doi)	Documentation only
02 → 03	BibTeX	`dedupe.bib`: deduplicated bibliography with unique IDs	Automated (duplicate check)
03 → 03b	CSV	`decisions.csv`: study_id, title, decision (include/exclude/uncertain), rationale, confidence, pass1, pass2, agreement	Automated (kappa computation)
03b → 04	Markdown	`analysis-type-decision.md`: pairwise or NMA, rationale, human approval	Documentation only
04 → 05	CSV	`manifest.csv`: study_id, pdf_path, retrieval_method (unpaywall/manual), retrieval_status	Automated (file existence check)
05 → 06	CSV	`extraction.csv`: study_id + all extraction fields defined in template, with confidence ratings	Automated (range/consistency checks)
06 → 07	Multiple	R output files (.rds, .csv): effect estimates, heterogeneity stats, forest plot data, funnel plot data	Automated (model convergence)
07 → 08	Quarto	IMRaD manuscript sections (.qmd), figures (.png), tables (.csv)	Automated (structure/word count)
08 → 09	JSON	`grade_assessment.json`: outcome, overall_rating, domain ratings (rob, inconsistency, indirectness, imprecision, pub_bias), rationale per domain	Documentation only

4.13. S4. Overclaim Detection Patterns

The automated overclaim detection system (Stage 09) scans generated manuscript text for the following 12 patterns:

#	Pattern	Description	Example Flagged Text
1	Causal language for non-randomized data	Causal verbs (caused, led to, resulted in) used for observational findings	"Smoking *caused* increased mortality" (in cohort study)
2	Overgeneralization beyond study population	Claims extending beyond the actual study population	"All patients with cancer should…" (when study was TNBC only)
3	Recommendations without evidence level	Treatment recommendations without stating evidence quality	"Clinicians should prescribe…" (without GRADE context)
4	Unsupported superlatives	"First," "only," "best," "most effective" without citation	"The *most effective* treatment…"
5	Certainty language for uncertain findings	"Clearly," "definitely," "proves" for findings with wide CIs	"This *clearly demonstrates*…"
6	Extrapolation to unexamined outcomes	Claims about outcomes not measured in included studies	Safety conclusions drawn from efficacy-only data
7	Ignoring heterogeneity	Definitive statements when $I^2 > 50\%$	"Treatment consistently improves…" (when I^2=72%)
8	P-value misinterpretation	Equating statistical significance with clinical importance	"Highly significant (p<0.001)" implying large effect
9	Absence of evidence as evidence of absence	Claiming no effect when studies were underpowered	"No difference in mortality was found" (2 trials, N=200)
10	Selective reporting language	Emphasizing favorable outcomes while minimizing harms	Extensive efficacy discussion with one-sentence safety
11	Temporal extrapolation	Long-term conclusions from short follow-up	"Long-term survival benefit" from 12-month data
12	Comparison with absent comparator	Claims about superiority over treatments not studied	"Superior to standard chemotherapy" (when not directly compared)

4.14. S5. Feature Comparison Methodology

The feature comparison in Table 2 was constructed as follows:

1. **Source identification**: For each tool, we identified the primary peer-reviewed publication, official documentation website, and most recent user guides (accessed January–March 2026).
2. **Feature classification**: Features were classified as **AI-assisted** (bold) when the tool uses machine learning or LLM automation for that stage, **supported** (plain text) when the tool provides templates, forms, or manual workflows, or **absent** (dash) when the feature is not available.
3. **Verification**: Two authors independently classified features for each tool. Disagreements were resolved by consulting the tool's documentation.
4. **Limitations**: This comparison reflects published capabilities and may not capture unreleased features, beta functionality, or recent updates. Commercial tools may have added capabilities since our assessment date.
5. **Validation evidence**: Validation claims are based on the strongest published evidence (peer-reviewed > preprint > internal reports > no published evidence).

Sources consulted:

- Covidence: Kellermeyer et al. 2018, JMLA; covidence.org documentation
- TrialMind: Wang et al. 2025, npj Digital Medicine
- otto-SR: Cao et al. 2025, medRxiv preprint; ottosr.com
- Nested Knowledge: Kallmes et al. 2025, Cochrane Evidence Synthesis and Methods; nested-knowledge.com
- DistillerSR: distillersr.com documentation (no primary peer-reviewed validation publication identified)

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